



SUPPORT VECTOR MACHINE (SVM)



Supervised Learning for Classification & Regression

SVM: THE CORE IDEA



Max margin → better generalization



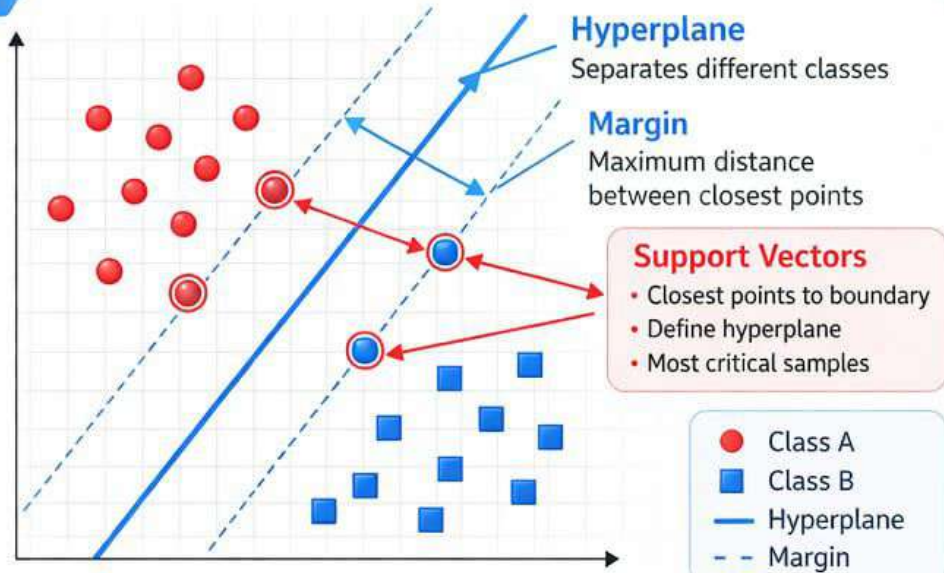
Only boundary points matter



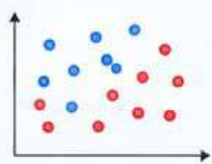
Insensitive to non-support points



Robust but sensitive to outliers near margin

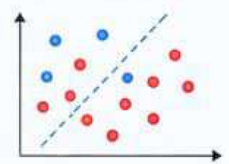


WORKING (STEP FLOW)



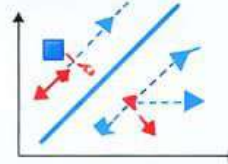
1. Plot Data

Visualize the given data points with their labels



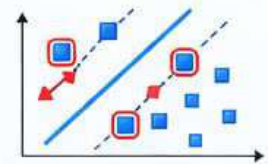
2. Find Boundary

Generate a candidate separating hyperplane



3. Maximize Margin

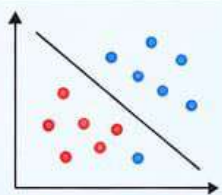
Adjust the hyperplane to find the largest separation gap



4. Use Support Vectors

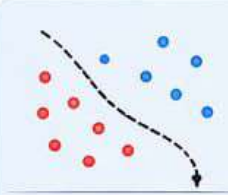
Identify key points that conclude the boundary

TYPES OF SVM



Linear SVM

Works for simple, linearly separable datasets

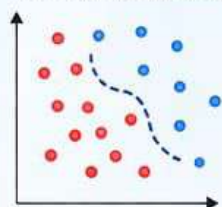


Non-Linear SVM

Uses kernels to handle complex, non-linear patterns

KERNEL TRICK

Input Space (2D)
Not linearly separable



Kernel Function
 $\phi(x)$

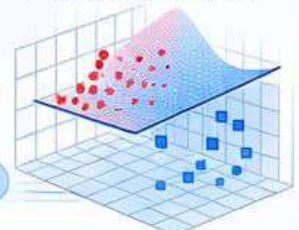
Maps to higher dimension

Linear

Polyomial

RBF

Feature Space (3D+)
Now linearly separable



The kernel trick transforms data into higher dimensions without explicit computation, enabling linear separation of complex patterns.

APPLICATIONS



Image Classification
Object detection & tagging



Spam Detection
Email filtering



Face Recognition
Biometric systems

KEY INSIGHTS



- ✓ Maximizes margin → better generalization
- ✓ Effective in high-dimensional spaces
- ✓ Works well with small datasets
- ✓ Kernel trick enables non-linear learning